Calculus mod 5

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About

These are some lecture notes for a series of talks given at the Chicago REU in the summer of 2026. I'm very grateful to Peter May, who organized the REU and let me speak. I also want to thank Kenta Suzuki; he and I run a blog 'hidden-phenomena', and lecture 2 of these notes is an expansion of an article from this blog.

Contents

Lecture 1

What is algebraic geometry?

1.1 Introduction

Algebraic geometry is the study of shapes defined by systems of polynomial equations.

What makes it interesting to single out the class of shapes defined by polynomial equations in particular? In other words – why is algebraic geometry its own subject, and not just a source of examples for algebraic topology and Riemannian geometry courses?

There are several answers one could give to this question. The point of view taken by this course is that these shapes are interesting because they make sense over many different number systems.

To illustrate what we mean, let's single out the parabola $y = x^2$. There are many different ways one could interpret it:

1. if x, y are interpreted as *real numbers*, we get the standard parabola shape in $\mathbb{R}^2$;

2. if x, y are *complex numbers*, then $y = x^2$ defines a certain 2-dimensional subset of the 4-dimensional space $\mathbb{C}^2$;

3. if x, y are interpreted as *integers modulo 3*, then the parabola is the 'shape' consisting of the three points $(0, 0), (1, 1)$, and $(2, 1)$;

4. if x, y are interepreted as *integers*, then the parabola is essentially the set of all perfect squares: it contains points like $(5, 25), (13, 169)$, and $(-3, 9)$, where the x-coordinate can be any integer, and y is always its square.

The miracle of algebraic geometry is that these different incarnations of the shape $y = x^2$ are somehow related. The purpose of algebraic geometry, from this point of view, is to develop a language with which one can talk about all of these different incarnations at once, and therefore reason about the relationships between them.

Example 1.1.1 (Integrals and Diophantine equations)**.** A high school calculus student can integrate

$$\int \frac{dx}{\sqrt{1-x^2}}. \tag{1.1.2}$$

However, the similar looking integral

$$\int \frac{dx}{\sqrt{x^3-2}} \tag{1.1.3}$$

would give even an expert undergraduate calculus student trouble!

When approaching the first integral, it is tempting to substitute $x = \sqrt{1-y^2}$. Then x and y obey the algebraic relationship

$$x^2 + y^2 = 1.$$

This allows you to evaluate (1.1.2) in terms of trigonometric functions (in fact, (1.1.2) evaluates to $\arcsin(y) + C$).

Similarly, the second integral is related to the algebraic curve

$$y^2 = x^3 - 2.$$

These two curves, $x^2+y^2 = 1$ and $y^2 = x^3-2$, are related to the Diophantine equations

$$a^2 + b^2 = c^2 \tag{1.1.4}$$

and

$$a^2c = b^3 - 2c^3. \tag{1.1.5}$$

It's very easy to find (with proof!) all the integer solutions to (1.1.4); this was done by the ancient Greek mathematician Euclid; and even before Euclid, the mathematician Diophantus found an infinite family of solutions, and the Babylonians found many particular solutions.

However, (1.1.5) is a much trickier Diophantine equation. It has essentially one solution (observe $25 = 27 - 2$), but proving this is quite tricky unless you know some algebraic number theory!

Perhaps remarkably, the geometry of $x^2 + y^2 = 1$ explains why (1.1.2) is easy to solve, and why (1.1.4) is easy to solve; and the geometry of $y^2 = x^3 - 2$ explains why (1.1.3) and (1.1.5) are both hard to solve.

Example 1.1.6 (Dimension and point counts)**.** Consider the parabola $y = x^2$. Over the real numbers, the graph of $y = x^2$ gives a 1-dimensional object. And in modulo n arithmetic, there are exactly n pairs (x, y) which solve

$$y \equiv x^2 \pmod{n},$$

as x can be any residue modulo n, and then y is determined.

Consider now the polynomial equation

$$x + y + z = 0.$$

Over the real numbers, this defines a 2-dimensional plane in 3-dimensional space. And in modulo n arithmetic, there are exactly n^2 solutions to $x + y + z \equiv 0 \pmod{n}$.

Note that, in both cases, the real number incarnation of our shape had dimension d, and the 'modulo n' incarnation of our shape was a set of size n^d. The geometric notion of dimension is thus reflected arithmetically by the size of the solution set in modular arithmetic.

Remark 1.1.7. Dimension, intuitively, measures the 'number of free parameters' in a shape. If we are selecting d independent numbers in $\mathbb{Z}/n\mathbb{Z}$, then there are n^d total ways to make our selection, which explains why n^d corresponds to d-dimensions.

Example 1.1.8 (A more complicated dimension and point count example)**.** Consider the equation

$$x^2 + y^2 = 1. \tag{1.1.9}$$

Over $\mathbb{R}$, the solutions to (1.1.9) form a circle – a 1-dimensional shape.

How many solutions are there modulo n? Define

$$N_n := \text{number of solutions to } x^2 + y^2 \equiv 1 \pmod{n}.$$

Determining N_n is a fun exercise in elementary number theory[1]. When $n = p^k$ is the power of a prime number p, like $n = 289 = 17^2$ or $n = 243 = 3^5$, then N_n is easy to determine:

$$N_{p^k} = \begin{cases} p^k - p^{k-1} & p \equiv 1 \pmod{4}, \\ p^k + p^{k-1} & p \equiv 3 \pmod{4}, \\ 2p^k & p = 2, k > 1 \end{cases}.$$

From this, using the Chinese remainder theorem, we can deduce that $N_n = O(n)$; here, we're using big-O notation, meaning that N_n grows roughly as fast as n (and in particular, it grows slower than n^2 or n^3). This $O(n^1)$ is again related to the fact that the circle is 1-dimensional.

Example 1.1.10 (Point counting on the sphere)**.** In the vein of the previous example, consider the equation

$$x^2 + y^2 + z^2 = 1.$$

Over the real numbers, this defines the unit sphere, a 2-dimensional shape. One can use elementary number theory, similarly to the previous example, to show that the number of solutions to

$$x^2 + y^2 + z^2 \equiv 1 \pmod{n}$$

is $O(n^2)$. For example, when $n = p^k$ where $p \equiv 1 \pmod{4}$ is a prime number, then there are $p^{2k} - p^{2k-1} = \left(1 - \frac{1}{p}\right) n^2$ solutions. The 2 in this $O(n^2)$ reflects the 2-dimensional nature of the solution set.

[1] Hint: Use Legendre symbols!

1.2 Prespaces

To help formalize these ideas, we introduce now the notion of a *prespace*. A prespace is to algebraic geometry as a point-set topological space is to geometric topology: that is, all the objects of (classical) algebraic geometry are prespaces, but in practice algebraic geometers usually do not study arbitrary prespaces, and instead study prespaces obeying many extra properties. Compare this to how a geometric topologist does not study arbitrary point-set topological spaces, but usually assumes their spaces are Hausdorff, or metrizable, or locally Euclidean, etc.

In the introduction, we explained that the main benefit of studying shapes cut out by systems of polynomial equations is how they are realized over many different number systems. We are going to define a *prespace* as a shape which has an incarnation over every possible number system, where by number system we will mean *ring*.

Warning 1.2.1. In these notes, and in most notes on algebraic geometry, a ring is always assumed to be *commutative and unital*.

Definition 1.2.2. A *prespace* is a functor

$$X : \mathrm{Ring} \to \mathrm{Set}\,.$$

Remark 1.2.3 (What is a functor?)**.** For those who do not know category theory, when we say $X : \mathrm{Ring} \to \mathrm{Set}$ is a *functor*, we mean that,

1. for every ring A, we can produce a *set* $X(A)$, *and*

2. for every ring *homomorphism* $A \to B$, we can produce a *function* of sets $X(f) : X(A) \to X(B)$,

such that

1. whenever $f : A \to B$ and $g : B \to C$ are two ring homomorphisms, we have $X(g \circ f) = X(g) \circ X(f)$,

2. for any ring A, the function $X(\mathrm{id}_A)$ is the identity function on $X(A)$.

Remark 1.2.4. It is useful to think of mathematical definitions as consisting of two parts: data, and then properties which that data must obey. For example, a group consists of the data of a set and a binary operation, obeying certain properties (like associativity).

Similarly, the data of a functor is the sets $X(A)$ and the functions $X(f)$, and the property this data must obey is $X(g \circ f) = X(g) \circ X(f)$.

Definition 1.2.5. A *morphism* of prespaces

$$\eta : X \to Y$$

is a natural transformation of functors.

Remark 1.2.6 (What is a natural transformation?)**.** A morphism of prespaces is the data of, for every ring A, a function

$$\eta(A) : X(A) \to Y(A),$$

such that, for any ring homomorphism $\varphi : A \to B$, the diagram

$$\begin{tikzcd} X(A) \arrow[r, "\eta(A)"] \arrow[d, "X(\varphi)"] & Y(A) \arrow[d, "Y(\varphi)"] \\ X(B) \arrow[r, "\eta(B)"] & Y(B) \end{tikzcd}$$

commutes; that is, $Y(\varphi) \circ \eta(A) = \eta(B) \circ X(\varphi)$.

Remark 1.2.7. The reader should check that the following statements:

1. if $\eta : X \to Y$ and $\theta : Y \to Z$ are morphisms of prespaces, then the map $\theta \circ \eta : X \to Z$ defined by $(\theta \circ \eta)(A) := \theta(A) \circ \eta(A)$ is also a natural transformation;
2. a morphism of prespaces $\eta : X \to Y$ has an inverse $\eta^{-1} : Y \to X$ if, and only if, for every ring A, the function $\eta(A)$ is a bijection of sets.

Example 1.2.8 (The parabola)**.** We define the *parabola* as the prespace

$$P : \text{Ring} \to \text{Set}$$

given by

$$P(A) := \{(x, y) \in A \times A \mid y = x^2\}.$$

Given a ring homomorphism $\varphi : A \to B$, we define

$$P(\varphi) : P(A) \to P(B)$$

as the function

$$P(\varphi)(x, y) = (\varphi(x), \varphi(y)).$$

This pair $(\varphi(x), \varphi(y))$ belongs to $P(B)$ because, whenever $y = x^2$, we have

$$\varphi(y) = \varphi(x^2) = \varphi(x \cdot x) = \varphi(x) \cdot \varphi(x) = \varphi(x)^2,$$

as φ is a ring homomorphism.

Example 1.2.9 (Affine space)**.** We define *affine n-space* as the prespace

$$\mathbb{A}^n : \text{Ring} \to \text{Set}$$

given by

$$\mathbb{A}^n(A) := \{(x_1, ..., x_n) \in A^n\}.$$

1.3 Affine schemes

We now single out a very special class of prespaces.

Definition 1.3.1. Let R be a ring. The *affine scheme* associated to R is the prespace

$$\mathrm{Spec}(R) : \mathrm{Ring} \to \mathrm{Set}$$

given by

$$\mathrm{Spec}(R)(A) := \{\text{ring homomorphisms } \varphi : R \to A\}.$$

Example 1.3.2. Consider the ring $\mathbb{Z}[x_1, ..., x_n]$ of all polynomials in the variables $x_1, ..., x_n$. Then the affine scheme $\mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n])$ coincides with the prespace $\mathbb{A}^n$ of Example 1.2.9; that is, there is an isomorphism of prespaces

$$\eta : \mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n]) \simeq \mathbb{A}^n.$$

This isomorphism η is defined as follows. For a ring A, set

$$\eta(A) : \mathrm{Spec}(\mathbb{Z}[x_1, ..., x_n])(A) \to \mathbb{A}^n(A)$$

the function sending a ring homomorphism $\varphi : \mathbb{Z}[x_1, ..., x_n] \to A$ to the n-tuple $(\varphi(x_1), ..., \varphi(x_n)) \in \mathbb{A}^n(A)$.

We leave it to the reader to check that η defines a morphism of prespaces. To see it is an isomorphism (in light of Remark 1.2.7), it is enough to check that $\eta(A)$ is always a bijection; we leave this to the reader (it is the universal property of polynomial rings).

Remark 1.3.3. The parabola P of Example 1.2.8 is also an affine scheme:

$$P = \mathrm{Spec}(\mathbb{Z}[x, y]/(y - x^2)).$$

Can you see why?

Remark 1.3.4. Can you describe the set of ring homomorphisms

$$\mathbb{Z}[x, y, z]/(x^2 + y^2 + z^2 - 1) \to A,$$

for A an arbitrary ring, concretely?

Lecture 2

Tangent vectors and Hensel's lemma

2.1 Introduction

Calculus studies *approximations* to *continuous* quantities. Number theory typically involves studying *exact* answers to questions about *discrete* quantities. These seem, therefore, to be unrelated fields.

Yet Kurt Hensel made a remarkable discovery, showing that one could use ideas from calculus to solve number theory problems. We are going to explain Hensel's wonderful discovery by trying to solve polynomial equations in modular arithmetic. More particularly, we are going to try and find all integers x such that

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3000}. \tag{2.1.1}$$

While this sort of problem – solving a polynomial equation in modular arithmetic – might seem a bit strange, let me say a few things about why one would solve this.

2.1.1 Divisibility

One can profitably view the evolution of number theory through the lens of Diophantine equations. The first Diophantine equation you might try to solve is one of the form

$$Ax = B,$$

like $5x = 10$ or $2x = 7$. The theory of *integer divisibility* gives a completely satisfactory understanding of this class of Diophantine equations: you can solve $Ax = B$ exactly when B is divisible by A, in which case there is a unique integer solution, and there's a pretty efficient algorithm (long division!) which can be used to check divisibility.

To ask for the 'next simplest' Diophantine equation, there are two directions we could go. One could either increase the *degree* of the Diophantine equation, or increase the number of variables.

2.1.2 Increasing the degree

If we increase degrees, we're led to the *rational root theorem*, which gives a very simple algorithm for enumerating all integer solutions to any polynomial equation of the form

$$A_d x^d + A_{d-1} x^{d-1} + \cdots + A_0 = 0,$$

where the A_i are integers. If you're only interested in integer solutions, the rational root theorem is actually pretty easy, and only requires you to know about divisibility.

Proposition 2.1.2. *Any integer solution to*

$$A_d x^d + \cdots + A_0 = 0$$

must be a divisor of A_0.

Proof. If n is an integer solving this equation, then

$$A_0 = -n \cdot (A_1 + A_2 n + \cdots + A_d n^{d-1}),$$

and so $n \mid A_0$. $\square$

To enumerate all the divisors of A_0, you can do trial divisions, so it's not so bad (especially if you make the observation that you need to only check for divisors up to $\sqrt{A_0}$, as divisors come in pairs!).

2.1.3 Increasing the number of variables

The other way we could have made our Diophantine equation harder would be to increase the number of variables instead of the degree; in other words, we could try to solve an equation like

$$Ax + By = C. \tag{2.1.3}$$

This is an algebraic geometry course, not a number theory course, so let me just say that solving (2.1.3) is more or less equivalent to understanding the theory of *unique prime factorization* and *greatest common denominators.* Thus, an essential phenomenon in number theory – unique prime factorization – can be discovered by pondering a Diophantine equation.

2.1.4 To our question

To me, the point of mathematics is to *discover interesting mathematical phenomena*. I do not care about Diophantine equations or other math problems in their own right; I care about phenomena. However, it has been proven again and again throughout history that humans are really quite bad at observing mathematical phenomena in a background. We somehow need to think about an interesting problem in order for it to provoke us to discover something. So, every Diophantine equation, or other math problem, that I encounter, I view as a tool, with which I can use to either look for new phenomena (when I'm trying to solve an open problem for my research), or which I can use to test my understanding of existing mathematical phenomena (when I'm solving an exercise). Diophantine equations have proven to be a great source of number theoretic phenomena, which is why people are interested in them.

The rational root theorem and unique prime factorization are both wonderful phenomena, but ultimately are not that complicated. Thus it's natural to increase the complexity even more: what if we have a Diophantine equation which both has large degree, and which has multiple variables? This is where things get much much harder!

The simplest source of such equations is to allow yourself to have two variables x and y, and allow your Diophantine equation to be as complicated as you want in x, but force it to be *linear* in y. For example,

$$x^2 + 1 = 5y. \tag{2.1.4}$$

Note, though, that finding integer solutions to (2.1.4) is equivalent to solving

$$x^2 + 1 \equiv 0 \pmod 5.$$

Using this trick, Diophantine equations which are *polynomial* in x and *linear* in y are always equivalent to solving polynomial equations in modular arithmetic.

So, a natural 'next step' in solving Diophantine equations is to ask how to solve equations of the form

$$p(x) \equiv 0 \pmod N,$$

where $p(x)$ is some polynomial and N is some integer.

We are going to illustrate, through (2.1.1), how you can go about solving equations of this type.

2.2 A first reduction

Let's go back to our starting example. The Chinese remainder theorem tells us that solving

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3000}$$

is equivalent to solving the *three* different congruences

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{2^3},$$

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3},$$

and

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{5^3}.$$

At first it seems we've made our problem *harder*, not easier: we started with one equation, and now we have three! However, there is a sense in which our life has been made simpler: to solve

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{2^3},$$

we need only plug in $x = 0, 1, 2, 3, 4, 5, 6,$ and 7. This is only eight values to try! If we simpliyfy our polynomial to

$$x^3 - 17x^2 + 12x + 16 \equiv x^3 - x^2 + 4x \pmod{8},$$

and plug in each of these values, then it's not that hard to check that the only solutions are

$$x \equiv 0, 4 \pmod{8}.$$

Similarly, we can check that the only solution to

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{3}$$

is $x \equiv 1 \pmod{3}$.

Just by plugging in a 11 values, we managed to solve 2 of our 3 equations. Unfortunately, plugging in every integer between 0 and 124 to solve our equation modulo 125 is a bit too hard for me. Fortunately, Hensel found a cleverer way to handle this.

2.3 5, 25, 125

If

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{5^3},$$

then $x^3 - 17x^2 + 12x + 16$ is divisible by 125. In particular, it is also divisible by 5, meaning

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{5}. \tag{2.3.1}$$

(2.3.1) is much easier to solve, as we need only plug in five possibilities. If you do this, you'll find that the only solution is $x \equiv 2 \pmod{5}$.

Observe that

$$5^3 - 17 \cdot 5^2 + 12 \cdot 5 + 16 = -20$$

is divisible by 5, but not by 125.

Hensel thought that being divisible by 5 made -20 *close* to being divisible by 125. After all, -20 is divisible by five once, and once is almost three times!

On its own, this thought would probably not be so helpful. What made this so remarkable was what Hensel did *next*. Before we can explain that, though, we take a brief digression to recall an insight from calculus.

2.4 Derivatives

In calculus, one typically defines the derivative as the limit

$$f'(x) = \lim_{\epsilon \to 0} \frac{f(x+\epsilon) - f(x)}{\epsilon}.$$

It is often more intuitive to think that $f'(x)$ is the unique real number having the property that

$$f(x+\epsilon) = f(x) + f'(x)\epsilon + o(\epsilon). \tag{2.4.1}$$

Here, $o(\epsilon)$ is *little-oh* notation, and we use it to denote a quantity which is significiantly smaller than ϵ, in the sense that $o(\epsilon)$ is an error term obeying $\lim_{\epsilon \to 0} o(\epsilon)/\epsilon = 0$. But you should see $o(\epsilon)$ and think "significantly smaller than ϵ."

The point of (2.4.1) is that a derivative is an *approximation* of a function.

Analysts devised limits and little-oh notation as a way of talking about approximations and very small numbers. The idea of approximation iso fundamental that it occurs even in algebra and algebraic geometry. However, algebraic geometers have a different way of formalizing this idea. This formalization is inferior at solving problems analysts care about, but it is very useful when solving problems algebraists care about.

2.4.1 The algebraic ϵ

Consider the ring $\mathbb{R}[\epsilon]/(\epsilon^2)$, which I will often denote by just $\mathbb{R}[\epsilon]$.

This means we take the real numbers $\mathbb{R}$, and freely adjoin a new number ϵ obeying $\epsilon^2 = 0$. That is, A is the ring whose elements are of the form $x + x' \cdot \epsilon$, where $x, x' \in \mathbb{R}$, and with addition and multiplication defined as

$$(x + x'\epsilon) + (y + y'\epsilon) := (x+y) + (x'+y')\epsilon,$$

$$(x + x'\epsilon)(y + y'\epsilon) := xy + (xy' + x'y)\epsilon.$$

We should think that A is a ring containing all the usual real numbers, together with this new very small number ϵ. Consider the cubic polynomial

$$f(x) = x^3 - 17x^2 + 12x + 16.$$

In a calculus class, we'd say it's derivative is

$$f'(x) = 3x^2 - 34x + 12.$$

Using our ring $\mathbb{R}[\epsilon]$, we can discover this derivative in a different way. In the ring $\mathbb{R}[\epsilon]$, observe that

$$\begin{aligned} f(x+\epsilon) &= (x+\epsilon)^3 - 17(x+\epsilon)^2 + 12(x+\epsilon) + 16 \\ &= (x^3 + 3x^2\epsilon) - 17(x^2 + 2x\epsilon) + 12(x+\epsilon) + 16 \\ &= f(x) + \epsilon \cdot (3x^2 - 34x + 12). \end{aligned}$$

In other words, in the ring $\mathbb{R}[\epsilon]$, we literally have the equality

$$f(x+\epsilon) = f(x) + \epsilon \cdot f'(x).$$

In fact, this works for *any* polynomial function! This ring $\mathbb{R}[\epsilon]$ is perfectly suited for talking about derivatives of polynomials.

2.5 Newton's method

We now remind the reader of one more topic in standard calculus: *Newton's method*. Fix $f : \mathbb{R} \to \mathbb{R}$ some differentiable function, and suppose we want to solve $f(x) = 0$. Newton's method starts with an *approximate* solution x_0 (so $f(x_0) \approx 0$), and then produces a slightly better approximation to a solution. If everything goes well, iterating Newton's method will converge to an exact solution.

Newton's method is based on a simple observation. We know that

$$f(x_0 + \epsilon) \approx f(x_0) + f'(x_0)\epsilon.$$

So, just solve for the value of ϵ making

$$f(x_0) + f'(x_0)\epsilon = 0.$$

Then, replace x_0 by the new approximate solution $x_0 + f'(x_0)\epsilon$.

Newton's method isn't perfect (as any numerical analyst will tell you), it is a pretty decent root finding method, and works well enough for our application.

2.6 Hensel's lemma

Now, on to Hensel's lemma. Recall our problem: we wanted to solve

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod{125}.$$

We knew that $x \equiv 2 \pmod 5$ solves

$$x^3 - 17x^2 + 12x + 16 \equiv 0 \pmod 5.$$

This is close to what we want, but not quite.

Following Hensel, we will try to upgrade our solution mod 5 to a solution modulo 25 (and then upgrade that solution to one which exists mod 125).

Hensel made a remarkable observation. From the point of view of modern algebraic geometry, Hensel realized the following: the relationship between $\mathbb{Z}/25$ and $\mathbb{Z}/5$ is analogous to the relationship between $\mathbb{R}[\epsilon]$ and $\mathbb{R}$. More precisely,

1. the ring $\mathbb{R}[\epsilon]$ contains the number ϵ which has $\epsilon^2 = 0$, and the ring $\mathbb{Z}/25$ contains the number 5 which has $5^2 = 0$;

2. their is a surjection ring homomorphism $\mathbb{R}[\epsilon] \to \mathbb{R}$ sending ϵ to 0, and there is a surjective ring homomorphism $\mathbb{Z}/25 \to \mathbb{Z}/5$ sending 5 to 0.

We now use this observation to run Newton's method in $\mathbb{Z}/25$. Set

$$p(x) := x^3 - 17x^2 + 12x + 16.$$

Observe that $p(2) = -20$. This is our 'approximate' solution to $p(x) \equiv 0 \pmod{25}$. In what sense is it an approximate solution? Well, just as ϵ's smallness was witnessed by the fact that $\epsilon^2 = 0$, in $\mathbb{Z}/25$ we have $(-20)^2 \equiv 0 \pmod{25}$.

Now, to do Newton's method, we should replace $p(2+\epsilon)$ with $p(2+5n)$, and attempt to solve for n, using multiples of 5 as our small numbers. Observe that

$$p(2+5n) \equiv p(2) + p'(2) \cdot 5n \pmod{25},$$

by the very same manipulations we used to prove $p(x+\epsilon) = p(x) + \epsilon p'(x)$ in $\mathbb{R}[\epsilon]$.

As

$$p'(x) = 3x^2 - 34x + 12,$$

we see that $p'(2) = -44$, so that

$$p(2+5n) \equiv -20 - 220n \equiv 5 + 5n \pmod{25}.$$

To solve $5 + 5n \equiv 0 \pmod{25}$, observe that it is equivalent to solve $1 + n \equiv 0 \pmod{5}$, which is pretty easy: $n \equiv 4 \pmod{5}$. Thus the solution to $p(x) \equiv 0 \pmod{25}$ is

$$x \equiv 2 + 5 \cdot 4 \equiv 22 \pmod{25}.$$

We're now close to solving our problem – all that remains is to upgrade our solution modulo 25 into a solution modulo 125. To do this, one can use Newton's method again:

$$p(22+25n) \equiv p(22) + p'(22) \cdot 25n \pmod{125}.$$

Observe that

$$p(22) \equiv 75 \pmod{125},$$
$$p'(22) \equiv 1 \pmod{5},$$

so that

$$p(22+25n) \equiv 75 + 25n \pmod{125},$$

and thus

$$x \equiv 22 + 25 \cdot (-3) \equiv 72 \pmod{125}$$

is a solution modulo 125. And so, using ideas from calculus, Hensel was able to solve a problem in number theory!

We can state this result more abstractly as follows, for those of you into that sort of thing.

Theorem 2.6.1 (Hensel's lemma)**.** *Let $f(x) \in \mathbb{Z}[x]$ be a polynomial, and p a prime number. Suppose $a \in \mathbb{Z}$ is an integer such that $f(a) \equiv 0 \pmod{p}$, and such that $f'(a) \not\equiv 0 \pmod{p}$.*

Then, for every integer $n \geq 1$, there is a unique *congruence class $b \in \mathbb{Z}/p^n$ such that*

1. $b \equiv a \pmod{p}$,
2. $f(b) \equiv 0 \pmod{p^n}$.

Remark 2.6.2. It's a good exercise to try generalizing this naive form of Hensel's lemma to accomodate situations where $f(a) \not\equiv 0 \pmod{p}$. You can ask me if you have questions, but try thinking about it on your own for a bit first!

Proof. We prove this by induction on n, with the base case of $n = 1$ being trivial. For the induction step, assume $b \in \mathbb{Z}/p^n$ is as desired, and we want to prove there exists a unique residue $b' \in \mathbb{Z}/p^{n+1}$ with the desired conditions. Choose some integer $b \in \mathbb{Z}$ whose image in $\mathbb{Z}/p^n$ is the residue class b; by abuse of notation, we do not distinguish between the integer and the residue class.

Observe that, for any such residue b', its image in $\mathbb{Z}/p^n$ obeys the two desired properties, and so must equal b. Thus any residue $b' \in \mathbb{Z}/p^{n+1}$ obeying our constraints must be of the form

$$b' = b + p^n \cdot k$$

for some integer k. Every residue of this form automatically obeys the $b' \equiv a \pmod{p}$ condition. And such a residue obeys $f(b') \equiv 0 \pmod{p^{n+1}}$ if and only if

$$f(b) + f'(b) \cdot p^n \cdot k \equiv 0 \pmod{p^{n+1}}.$$

As $f(b) \equiv 0 \pmod{p^n}$, we may write $f(b) = p^n \cdot j$ for some integer j.

Observe that $f'(b) \equiv f'(a) \pmod{p}$ (as $b \equiv a \pmod{p}$), and so $f'(b)p^n \equiv f'(a)p^n \pmod{p^{n+1}}$. Thus $b' = b + p^n \cdot k$ is as desired if and only if k obeys

$$p^n \cdot (j + f'(a)k) \equiv 0 \pmod{p^{n+1}},$$

or equivalently if

$$j + f'(a)k \equiv 0 \pmod{p}.$$

As $f'(a) \not\equiv 0 \pmod{p}$, there is a unique residue $k \in \mathbb{Z}/p$ solving this equation, and hence a unique $b' \in \mathbb{Z}/p^{n+1}$ with the desired properties. $\square$

2.7 Tangent vectors

In a calculus course, the derivative is also used to find tangents. We now explain how to think about tangent vectors in the language of algebraic geometry. For this, it's useful to focus on a concrete geometric example.

Consider the circle $x^2 + y^2 = 1$, which we will view as the prespace

$$C : \text{Ring} \to \text{Set},$$

$$C(\mathbb{R}[\epsilon]) := \{(x, y) \in A^2 \mid x^2 + y^2 = 1\}.$$

Remark 2.7.1. This prespace C is an affine scheme; in fact, $C = \operatorname{Spec} \mathbb{Z}[x, y]/(x^2 + y^2 = 1)$.

The set $C(A)$ contains all the points of $C(\mathbb{R})$. For example, $(\sqrt{2}/2, \sqrt{2}/2) \in C(A)$, and $(1, 0) \in C(A)$. But $C(A)$ also contains some new interesting points: note that $(1, \epsilon) \in C(A)$, because

$$1^2 + \epsilon^2 = 1 + 0 = 1.$$

What is the set $C(A)$? In other words, what are the quadruples of real numbers $(x, x', y, y') \in \mathbb{R}^4$ such that

$$(x + x'\epsilon)^2 + (y + y'\epsilon)^2 = 1? \tag{2.7.2}$$

The left hand side of (2.7.2) expands out to

$$(x^2 + 2xx'\epsilon) + (y^2 + 2yy'\epsilon) = (x^2 + y^2) + (2xx' + 2yy')\epsilon.$$

Thus, $C(A)$ can be described as

$$C(A) = \{(x + x'\epsilon, y + y'\epsilon) \mid x^2 + y^2 = 1, 2xx' + 2yy' = 0\}. \tag{2.7.3}$$

Remark 2.7.4. The tangent to the circle at the point (x_0, y_0) is the line

$$xx_0 + yy_0 = 1,$$

as you can compute from basic calculus. Does this remind you of (2.7.3)?

While one can profitably think about tangent vectors using tangent lines and tangent planes, there is another way to conceptualize them in algebraic geometry. The set $C(\mathbb{R}[\epsilon])$ contains the point

$$P' = \left(\frac{1}{2} - \sqrt{3} \cdot \epsilon, \frac{\sqrt{3}}{2} + \epsilon\right).$$

Intuitively, we think that this point is a small deformation of $P = (1/2, \sqrt{3}/2) \in C(\mathbb{R})$. The displacement from P to P' is thought of as an infinitesimally short vector; that infinitesimal displacement is a tangent vector, to an algebraic geometer. In other words, a tangent vector is an *infinitesimally small step which keeps you on the shape.*

2.7.1 The tangent space

Inspired by the example of $\mathbb{R}[\epsilon]$, we make the following definition.

Definition 2.7.5. Let $X : \text{Ring} \to \text{Set}$ be a prespace. The *tangent space* of X is the prespace

$$TX : \text{Ring} \to \text{Set}$$

defined by

$$TX(A) := X(A[\epsilon]/(\epsilon^2)).$$

Remark 2.7.6. For any ring A, there are natural maps

$$A \hookrightarrow A[\epsilon] \xrightarrow{\epsilon=0} A.$$

The first map is injective, and the second map is surjective (it sends $a_0 + a_1\epsilon$ to a_0).

These maps give rise to functions

$$X(A) \to X(A[\epsilon]) \to X(A),$$

which in turn give us natural transformations

$$X \to TX \to X.$$

What are these two maps? The second one, $TX \to X$, sends a tangent vector to its basepoint. The first map, $X \to TX$, is often called the *zero section*: it sends a point to the "0 vector" at that point.

Lecture 3

Formally etale morphisms

3.1 Introduction

In last Thursday's lecture, we saw Hensel's lemma. Today, we are going to revisit it, and start trying to generalize it to systems of polynomial equations in several variables, as opposed to a single equation in one variable. Like last week, today is mostly theory of prespaces, and Thursday is mostly 'problem solving.'

To preface, let me say that, in a usual calculus class, the *tangent vector* is the fundamental notion. In algebraic geometry, as we saw in Hensel's lemma, calculus has a slightly different flavor: the fundamental object is the *square-zero extension*, a term I'll define today, and the goal is to study how to 'extend points along square-zero extensions.' I'll say what that means precisely later today, and then on Thursday we'll study this problem.

3.2 Ideals

As I said at the start, algebraic geometry is the study of shapes defined by systems of polynomial equations. Let me make precise what exactly I mean by 'system of equations' today.

To start, let's think about shapes cut out in $\mathbb{R}^2$. The parabola is cut out by

$$y = x^2,$$

and the point $(1, 4)$ can be cut out by the system of equations

$$x = 1,$$

$$y = 4.$$

It's slightly convenient to normalize these so that all our equations are of the form

$$\text{quantity} = 0,$$

so let's think of the parabola as solving

$$y - x^2 = 0,$$

and let's think of the point $(1, 4)$ being the solution to

$$x - 1 = 0,$$

$$y - 4 = 0.$$

The advantage to writing things this way is we can now encode systems of equations as lists: the parabola is

$$(y - x^2),$$

and the system defining $(1, 4)$ is

$$(x - 1, y - 4).$$

The point $(1, 4)$ can also be defined as the system of equations $(x+y-5, 2x+y-6)$. Moreover, the parabola can also be defined by the system of equations $(y - x^2, 5(y - x^2), x(y - x^2))$.

We now introduce the notion of an *ideal*. You should think that an ideal is a system of equations which is "as large as possible" without changing the solution set, so as to avoid the ambiguities encountered above.

Definition 3.2.1. An *ideal* of a ring A is a subset $I \subseteq A$ such that

1. for $i, j \in I$, we have $i + j \in I$;
2. and for $a \in A, i \in I$, we have $ai \in I$.

Remark 3.2.2. Depending on what you mean by solution set, you might think that a *radical ideal* is the largest possible system of equations you obtain without changing the solution set. We will not go there.

Definition 3.2.3. Let A be a ring. For $a_1, ..., a_n \in A$, the *ideal generated by* $a_1, ..., a_n$, denoted by

$$(a_1, ..., a_n) \subseteq A,$$

is the smallest ideal containing each a_i.

Remark 3.2.4. As an arbitrary intersection of ideals is an ideal (you should check this!), there is always a unique smallest ideal containing $a_1, ..., a_n$.

Ideation 3.2.1 (Closed immersions)**.** Let A be a ring, and I an ideal. What is the closed subspace of $\mathrm{Spec}(A)$ cut out by I?

Let $Z \hookrightarrow \mathrm{Spec}(A)$ be the closed immersion cut out by I.

For a ring R, we should then get some injective map

$$Z(R) \hookrightarrow \mathrm{Hom}(A, R).$$

So, we want to specify some subset of $\mathrm{Hom}(A, R)$, which corresponds "the points belonging to the locus where the elements of I vanish."

Think of an example like $A = \mathbb{Z}[x, y], I = (y - x^2)$. Then a ring homomorphism

$$\mathbb{Z}[x, y] \to R$$

is the same as giving two numbers $r_1, r_2 \in R$. We think of r_1, r_2 as being the x- and y-coordinates of a point on the 2-dimensional plane. What does it mean for that point to belong to the parabola? Well, just that $r_2 = r_1^2$. This suggests that

$$Z(R) = \{\varphi : A \to R \mid \varphi(I) = 0\},$$

or equivalently

$$Z(R) = \mathrm{Hom}(A/I, R).$$

Thus $Z = \mathrm{Spec}(A/I)$.

Definition 3.2.5. A *basic closed immersion* is a morphism of prespaces of the form

$$\mathrm{Spec}(A/I) \to \mathrm{Spec}(A).$$

3.3 More on closed immersions

The notion of closed immersion can be confusing at first, so I want to really hammer it in by talking about why we can view elements of a ring A as functions on $\mathrm{Spec}(A)$.

Let's start by recalling affine space.

Definition 3.3.1. The *n-dimensional affine space* is the affine scheme

$$\mathbb{A}^n := \mathrm{Spec}\, \mathbb{Z}[x_1, ..., x_n].$$

Remark 3.3.2. For every ring A, we have

$$\mathbb{A}^n(A) := \{(a_1, ..., a_n) \in A^n\}.$$

Definition 3.3.3 (Functions on a prespace)**.** A *regular function* on a prespace X is a morphism of prespaces

$$f : X \to \mathbb{A}^1.$$

Remark 3.3.4. Recall this means that, for every ring A, we have a function of sets

$$X(A) \to A.$$

That is, for every A-valued point of X, we output a number in A.

Proposition 3.3.5. *Let $X = \mathrm{Spec}(A)$. Then regular functions on X are in bijection with elements of A.*

Proof. Let $a \in A$. Define

$$a : \operatorname{Spec}(A) \to \mathbb{A}^1$$

as the morphism which on R-points acts as

$$\operatorname{Hom}(A, R) \to R$$

$$\varphi \mapsto \varphi(a).$$

Thus every element of A gives rise to a regular function on X. Conversely, let

$$\eta : \operatorname{Spec}(A) \to \mathbb{A}^1$$

be a regular function. Set $a = \eta(A)(\mathrm{id}_A)$. □

Remark 3.3.6. Historically, one big motivation for algebraic geometry was that a compact metric space X can be recovered from the ring $C(X, \mathbb{R})$ of continuous functions $X \to \mathbb{R}$. Thus one can think of rings and shapes as being the same sorts of objects: each shape uniquely gives rise to a ring of functions, and conversely (via the spectrum) each ring gives rise to a shape.

3.4 Basic open immersions

There is one more type of basic map which will show up in our statement of Hensel's lemma, so I want to explain it.

An open set is the complement of a closed set. In algebraic geometry, how do we make the complement of a set? Thinking as in Remark 3.3.6, one way you can form the complement is to *add new functions* to your ring.

For example, on the affine line $\mathbb{A}^1$, you have all polynomial functions: $\mathbb{A}^1 = \operatorname{Spec} \mathbb{Z}[x]$. But if you take the complement of the closed set $\{0\}$, then you can also evaluate the function $1/x$.

On $\mathbb{A}^1$, the function $1/x$ is undefined, because x could equal 0. But on $\mathbb{A}^1 \setminus \{0\}$, the function $1/x$ is perfectly valid. Thus

$$\mathbb{A}^1 \setminus \{0\} = \operatorname{Spec} \mathbb{Z}[x, 1/x].$$

Similarly, we make the following definition.

Definition 3.4.1. The *complementary open immersion* to the closed immersion

$$\operatorname{Spec}(A/a) \hookrightarrow \operatorname{Spec}(A)$$

is defined as the morphism of prespaces

$$\operatorname{Spec}(A_a) \hookrightarrow \operatorname{Spec}(A).$$

3.5 The Hensel analogy

Recall that Hensel's lemma was based on the analogy between $\mathbb{R}[\epsilon] \xrightarrow{\epsilon=0} \mathbb{R}$ and $\mathbb{Z}/25 \to \mathbb{Z}/5$. We now define a general class of ring homomorphisms fitting into this analogy.

Definition 3.5.1. A *square-zero extension* is a ring homomorphism $\varphi : A \to B$ such that

1. φ is surjective,
2. for any $a_1, a_2 \in A$ such that $\varphi(a_1) = \varphi(a_2) = 0$, we have $a_1 a_2 = 0$.

Remark 3.5.2. If we set $I := \ker \varphi$, then the second condition can be written as $I^2 = 0$. In particular, every $a \in A$ with $\varphi(a) = 0$ has $a^2 = 0$.

We can now restate Hensel's lemma from last time.

Theorem 3.5.3 (Hensel's lemma)**.** *Let $\varphi : A \to B$ be a square-zero extension of rings. Suppose $f(x) \in \mathbb{Z}[x]$. Set*

$$X = \operatorname{Spec} \mathbb{Z} \left[x, \frac{1}{f'(x)} \right] /(f(x)).$$

Then the function $X(\varphi) : X(A) \to X(B)$ is a bijection.

Proof. Set $I = \ker \varphi$, so that $B = A/I$.

An element of $X(B)$ is the same thing as a ring homomorphism

$$\mathbb{Z} \left[x, \frac{1}{f'(x)} \right] /(f(x)) \to A/I,$$

or equivalently some element $\bar{a} \in A/I$ such that

$$f(\bar{a}) = 0$$

and such that $f'(\bar{a})$ is a unit in A/I.

Choose $a \in A$ a lift of $\bar{a}$ to A/I. As $f(\bar{a}) = 0$, we have $f(a) \in I$. Moreover, as $f'(\bar{a})$ is a unit in A/I, we can find some $b \in A$ such that

$$b \cdot f'(a) = 1 + j$$

for some $j \in I$. Set $i := -bf(a) \in I$. As $i^2 = 0$, we have

$$f(a + i) = f(a) + f'(a) \cdot i.$$

But note that

$$f'(a) \cdot i = -bf(a)f'(a) = -f(a) \cdot (1 + j) = -f(a),$$

as $f(a) \cdot j \in I^2 = 0$. Thus $f(a + i) = 0$.

Hence $X(A) \to X(B)$ is surjective. For injectivity, note that $i' \in I$ has $f(a+i') = 0$ if and only if

$$f'(a)i' = -f(a),$$

and if this happens then

$$i' = i' + i'j = -bf(a),$$

so that our choice of i above is unique. □

Definition 3.5.4. A prespace X : Ring $\to$ Set is *formally etale* if, for any square-zero extension $A \to A/I$, the induced map

$$X(A) \to X(A/I)$$

is a bijection.

Remark 3.5.5. Thus Hensel's lemma tells us that every prespace $\operatorname{Spec} \mathbb{Z}[x, 1/f'(x)]/(f(x))$ is formally etale.

Lecture 4

High-dimensional Hensel's lemma

4.1 Introduction

Last time, we rephrased Hensel's lemma using square-zero extensions. Today, we're going to try and study how Hensel's lemma changes if you move beyond dimension 0 – so instead of one equation in one variable (which defines a $1-1=0$ dimensional solution space), we will look at systems of n equations in k variables (which, generically, defines a shape of dimension $n-k$).

As an example, consider

$$X = \operatorname{Spec} \mathbb{Z}[x, y, z]/(x^2 + y^2 + z^2 - 1).$$

Our goal today is to study the $\mathbb{Z}/p^2$-points of X.

Note that $X(\mathbb{R})$ is the unit sphere, sitting inside of 3-dimensional space.

4.2 Newton's method for hypersurfaces

In our proof of Hensel's lemma, we used Newton's method. To generalize Hensel's lemma, we will use a generalization of Newton's method. Imagine we have a point $(x_0, y_0, z_0) \in \mathbb{R}^3$ which 'approximately' belongs to the unit sphere. How can we make a point which is closer to living on the unit sphere?

Fortunately for us, multivariate Newton's method works pretty well: we just replace the derivative by the gradient. More precisely, set

$$f(x, y, z) = x^2 + y^2 + z^2.$$

We want to solve $f(x, y, z) = 1$. We have an approximate solution: $f(x_0, y_0, z_0) \approx 1$, and we want to perturb it to a genuine solution. Calculus tells us that

$$f(x_0+\epsilon_x, y_0+\epsilon_y, z_0+\epsilon_z) \approx f(x_0, y_0, z_0)+\frac{\partial f}{\partial x}(x_0, y_0, z_0)\cdot\epsilon_x+\frac{\partial f}{\partial y}(x_0, y_0, z_0)\cdot\epsilon_y+\frac{\partial f}{\partial z}(x_0, y_0, z_0)\cdot\epsilon_z.$$

Thus, we should choose our perturbation $(\epsilon_x, \epsilon_y, \epsilon_z)$ to obey

$$2x_0\epsilon_x + 2y_0\epsilon_y + 2z_0\epsilon_z = 1 - f(x_0, y_0, z_0). \tag{4.2.1}$$

We remark something interesting here: with Newton's method, we had a linear equation in one variable, and hence got a unique solution, but now we have a linear equation in *three* variables, and hence expect to have a 2-dimensional solution space.

This reflects some extra flexibility which we will have when Hensel lifting points, as we will see concretely next.

4.3 Lifting points on the sphere

Consider the point $(1, 1, 2) \in X(\mathbb{Z}/5)$, recalling that

$$1^2 + 1^2 + 2^2 \equiv 1 \pmod 5.$$

Unfortunately,

$$1^2 + 1^2 + 2^2 \equiv 6 \pmod{25},$$

so $(1, 1, 2) \notin X(\mathbb{Z}/25)$. However, it 'approximately'

To try and lift $(1, 1, 2)$ to $\mathbb{Z}/25$, we use implicit differentiation: a pertubration of this point looks like $(1 + 5a, 1 + 5b, 2 + 5c)$, and the same algebra from above (see (4.2.1)) shows that this lies in $X(\mathbb{Z}/25)$ if and only if

$$2(1 \cdot 5a + 1 \cdot 5b + 2 \cdot 5c) \equiv 1 - 6 \equiv -5 \pmod{25},$$

or equivalently if

$$a + b + 2c \equiv 2 \pmod 5. \tag{4.3.1}$$

There are 25 solutions to this congruence modulo 5; this gives us 25 different lifts of $(1, 1, 2)$ to a point of $X(\mathbb{Z}/25)$. For example, the solution

$$1 + 1 + 2 \cdot 0 \equiv 2 \pmod 5$$

to (4.3.1) gives rise to the point $(6, 6, 2) \in X(\mathbb{Z}/25)$. And indeed, observe

$$6^2 + 6^2 + 2^2 \equiv 76 \equiv 1 \pmod{25},$$

just as predicted!

4.4 A system of equations

Let's now upgrade from one equation in several variables to a system of equations in several variables. More precisely, let's consider

$$Y = \operatorname{Spec} \mathbb{Z}[x, y, z]/(x^2 + y^2 + z^2 - 1, (x - 1)^2 + y^2 + z^2 - 1),$$

corresponding to the system

$$x^2 + y^2 + z^2 = 1,$$
$$(x - 1)^2 + y^2 + z^2 = 1.$$

Over $\mathbb{R}$, this system looks like two spheres intersecting in a circle.

4.4.1 Newton's method

Let (x_0, y_0, z_0) be a point which approximately belongs to $Y(\mathbb{R})$. We can try using Newton's method to get a better approximation; this is very similar to before, but now our perturbation obeys two constraints instead of one. To spell things out, set

$$f(x,y,z) := x^2 + y^2 + z^2 - 1,$$

$$g(x,y,z) := (x-1)^2 + y^2 + z^2 - 1.$$

Then $(x_0 + \epsilon_x, y_0 + \epsilon_y, z_0 + \epsilon_z)$ should obey both

$$2x_0\epsilon_x + 2y_0\epsilon_y + 2z_0\epsilon_z = -f(x_0, y_0, z_0)$$

and

$$2(x_0 - 1)\epsilon_x + 2y_0\epsilon_y + 2z_0\epsilon_z = -g(x_0, y_0, z_0).$$

This cuts down the space of possible perturbations!

4.4.2 Solutions modulo 25

Consider the point $(3, 1, 1) \in Y(\mathbb{Z}/5)$. Let's try to lift this to a solution modulo 25. This requires us to solve the system

$$2(3 \cdot 5a + 1 \cdot 5b + 1 \cdot 5c) \equiv -10 \pmod{25},$$

$$2((3-1) \cdot 5a + 1 \cdot 5b + 1 \cdot 5c) \equiv -5 \pmod{25}.$$

This system is equivalent to solving the system

$$3a + b + c \equiv 4 \pmod 5,$$

$$2a + b + c \equiv 2 \pmod 5.$$

This system has *five* solutions modulo 5; one example solution is $a = 2, b = 1, c = 2$, leading to the point

$$(13, 6, 11) \in Y(\mathbb{Z}/25),$$

and indeed we can check

$$13^2 + 6^2 + 11^2 \equiv 326 \equiv 1 \pmod{25},$$

$$(13-1)^2 + 6^2 + 11^2 \equiv 301 \equiv 1 \pmod{25},$$

as expected.

4.5 Smoothness and the Jacobian criterion

If we abstract away the ideas from the previous section, we get the following variant of Hensel's lemma.

Theorem 4.5.1 (High dimensional Hensel's lemma)**.** *Let $f_1, ..., f_r \in \mathbb{Z}[x_1, ..., x_d]$ be polynomials, and let p be a prime number. Suppose that $(a_1, ..., a_d) \in (\mathbb{Z}/p)^d$ solves*

$$f_i(a_1, ..., a_d) \equiv 0 \pmod{p}$$

for each $i \in \{1, ..., r\}$, and that $d > r$ (so we have more variables than equations).

Consider the $d \times r$ matrix J with entries

$$J_{ij} := \frac{\partial f_i}{\partial x_j}(a_1, ..., a_d).$$

Note that these entries belong to $\mathbb{Z}/p$.

Assume that the matrix J has rank r. Then, for every $n \geq 1$, there exist exactly $p^{(n-1)(d-r)}$ points $(b_1, ..., b_d) \in (\mathbb{Z}/p^n)^d$ such that

1. *$b_i \equiv a_i \pmod{p}$ for each i,*
2. *$f_i(b_1, ..., b_d) \equiv 0 \pmod{p^n}$ for each i.*

Proof. Left as an exercise to the reader; it's quite instructive. $\square$

Lecture 5

Derivations

5.1 Introduction

This is our penultimate lecture.

In the first week, we saw how algebraic geometry was about one shape can have incarnations over different number systems, which spoke to each other.

In the second week, we saw how to algebraically capture closed and open immersions, and then continued exploring calculus in algebraic geometry.

This week, we're going to finish our exploration of calculus. We will begin with derivations.

5.2 A digression: modules

We start by reminding the reader of a helpful concept from commutative algebra.

Definition 5.2.1 (Modules)**.** A *module* over a ring R is analogous to a vector space over a field. Let me not beat you with a long list of axioms, but instead give you a feel: an R-module M is an abelian group together with a way of scaling elements of M by elements of R. So,

1. R^n is an R-module, where you add and scale termwise:

$$(r_1, ..., r_n) + (r'_1, ..., r'_n) = (r_1 + r'_1, ..., r_n + r'_n),$$

$$\lambda \cdot (r_1, ..., r_n) = (\lambda r_1, ..., \lambda r_n).$$

2. if R is a ring, and I is any ideal of R, then I is an R-module; for example, the set $5\mathbb{Z}$ (of multiples of 5) is a $\mathbb{Z}$-module, because we can rescale multiples of 5 by integers to get new multiples of 5, and we can add multiples of 5 together to get mutliples of 5.

5.3 Derivations

To more systematically treat the liftings done in the high-dimensional Hensel's lemma from last time, algebraic geometers introduced *derivations*.

Take $\pi : A \to A/I$ a square-zero extension; recall from last time that you should have in mind situations like $\mathbb{Z}/p^2 \to \mathbb{Z}/p$ or $\mathbb{R}[\epsilon] \to \mathbb{R}$, and that you should think of square-zero extensions as analogous to tangent vectors.

Fix X a general prespace. Last time, we introduced the notion of *formally etale* prespaces; these were prespaces for which the map

$$X(\pi) : X(A) \to X(A/I)$$

was always a bijection.

We now ask ourselves: what can we say about $X(\varphi)$ in general? That is,

1. when is $X(\pi)$ *surjective*?
2. when is $X(\pi)$ *injective*? when $X(\pi)$ is not injective, how large are its fibers?

The high dimensional Hensel's lemma, for example, showed us a situation where $X(\pi)$ was surjective, and had fibers of predictable sizes (and, as we saw, these fibers are easy to describe).

To answer this question in general, one needs something called the *cotangent complex*. This is a rather delicate device, so we will instead only explore a rather special case of this question – but a special case which is useful in practice.

So, for today we will assume that $X = \mathrm{Spec}(R)$ is an affine scheme. Then

$$X(A) = \mathrm{Hom}(R, A),$$
$$X(A/I) = \mathrm{Hom}(R, A/I).$$

5.4 Injectivity

What do the fibers of

$$X(A) = \mathrm{Hom}(R, A) \to \mathrm{Hom}(R, A/I) = X(A/I)$$

look like?

To start, let's fix an element $\bar{\varphi} : R \to A/I$, and try to understand all possible ring homomorphisms $\varphi : R \to A$ such that φ reduces to $\bar{\varphi}$ modulo I.

Example 5.4.1. Suppose $A \to A/I$ is the square-zero extension $\mathbb{Z}/25 \to \mathbb{Z}/5$, and

$$R = \mathrm{Spec}\,\mathbb{Z}[x, y]/(y - x^2)$$

is the parabola. Suppose $\bar{\varphi} : R \to A/I$ is the point $(3, 4)$, recalling tha

$$3^2 \equiv 4 \pmod 5.$$

Then the fiber of $X(\mathbb{Z}/25) \to X(\mathbb{Z}/5)$ above $\bar{\varphi}$ is the set

$$\{(3, 9), (8, 14), (13, 19), (18, 24), (23, 4)\}.$$

It is possible that this fiber is empty; let's ignore the question of surjectivity for now, and assume that the fiber has at least one element $\varphi : R \to A$. In this case, how many other elements can there be?

Imagine $\varphi' : R \to A$ is another point of the fiber. So, φ and φ' are two ring homomorphisms which both reduce to $\bar{\varphi}$ modulo I. In other words, $\varphi(r) - \varphi'(r)$ belongs to I for every $r \in R$.

Define

$$d : R \to I$$

as $d(r) := \varphi(r) - \varphi'(r)$. Observe that $d(r + r') = d(r) + d(r')$, and

$$\begin{aligned} d(rr') &= \varphi(rr') - \varphi'(rr') \\ &= \varphi(r)\varphi(r') - \varphi'(r)\varphi'(r') \\ &= \varphi(r)(\varphi(r') - \varphi'(r')) + \varphi'(r')(\varphi(r) - \varphi'(r)) \\ &= \varphi(r) \cdot d(r') + \varphi'(r') \cdot d(r) \\ &= \varphi(r) \cdot d(r') + (\varphi(r') - d(r')) \cdot d(r) \\ &= \varphi(r) \cdot d(r') + \varphi(r') \cdot d(r). \end{aligned}$$

In the penultimate step, we used that $I^2 = 0$, so that $d(r)d(r') = 0$.

Thus, d obeys the Leibniz rule! This inspires the following definition.

Definition 5.4.2 (Derivations)**.** Let R be a ring, and M an R-module. An *M-valued derivation* is a map

$$d : R \to M$$

such that

$$d(r + r') = d(r) + d(r')$$

and

$$d(rr') = rd(r') + r'd(r).$$

We write $\mathrm{Der}(R, M)$ for the set of all derivations $d : R \to M$.

Remark 5.4.3. When we have a square-zero extension $A \to A/I$, and a map $\bar{\varphi} : R \to A/I$, we can view I as an R-module in the following natural way: because $I^2 = 0$, for any $a, a' \in A$ such that $a \equiv a' \pmod{I}$ and any $i \in I$, we have

$$ai = a'i.$$

Thus I is naturally an A/I-module, and so by restriction of scalars along $\bar{\varphi}$, it is naturally an R-module as well:

$$r \cdot i := \bar{\varphi}(r) \cdot i.$$

Thus our maps $d : R \to I$ above are derivations in the sense of Definition 5.4.2.

Derivations answer our injectivity problem.

Theorem 5.4.4. *Let $X = \operatorname{Spec}(R)$ be an affine scheme, and $\pi : A \to A/I$ a square-zero extension. Consider some $\bar{\varphi} \in X(A/I)$. Then, either*

1. *the set*
$$F := \{\varphi \in X(A) \mid X(\pi)(\varphi) = \bar{\varphi}\}$$
is empty; or

2. *F contains an element φ, and there is a bijection of sets*
$$F \simeq \operatorname{Der}(R, I),$$
with the bijection sending a derivation d to the ring homomorphism $\varphi + d$.

Remark 5.4.5 (Torsors)**.** In the second situation, F is an example of a *torsor*: an object which becomes an abelian group when you declare some element φ its identity, but which has no 'natural' choice of identity.

Height is an example of a torsor: for any two points on Earth, you can discuss how much the difference of their heights, but there is no absolute height unless you choose a reference point (for example, people commonly discuss height above sea level). If you like physics, *potential energy* is a fancier example of height which is also a good torsor.

The set $\operatorname{Der}(R, M)$ has a natural group structure, but F does not: it is only after choosing a point φ that we can make it isomorphic to $\operatorname{Der}(R, M)$. You should think of the choice of φ as analogous to picking sea level.

Proof. This is essentially immediate from the above discussion: if $\varphi' \in F$, then $\varphi - \varphi'$ is a derivation, and conversely it's easy to see that if d is a derivation, then $\varphi + d \in F$. $\square$

5.4.1 Derivations are tangent vectors

Derivations are the algebraic geometry version of tangent vectors. To help this claim feel plausible, let's just work out derivations in one specific case.

Example 5.4.6. Set $X = \operatorname{Spec} \mathbb{Z}[x, y]/(x^2 + y^2 - 1)$ the circle. Let
$$A = \mathbb{R}[\epsilon], A/I = \mathbb{R}.$$

Recall that

1. $X(A/I) = X(\mathbb{R})$ is a pair $(x, y) \in \mathbb{R}^2$ such that $x^2 + y^2 = 1$,
2. $X(A)$ is a pair $(x + x'\epsilon, y + y'\epsilon) \in A^2$ such that $(x + x'\epsilon)^2 + (y + y'\epsilon)^2 = 1$.

In particular, $X(A) \to X(A/I)$ is *surjective*, because if (x, y) lies on the unit circle, where x, y are real numbers, then $(x + 0\epsilon, y + 0\epsilon) \in X(A)$ lies above (x, y) in $X(A) \to X(A/I)$.

The fiber of $X(A) \to X(A/I)$ above (x, y) is the set of all pairs
$$(x + x'\epsilon, y + y'\epsilon)$$

such that $(x+x'\epsilon)^2+(y+y'\epsilon)^2 = 1$. Using that (x, y) is fixed and that $x^2+y^2 = 1$, this is the same as the set of all pairs $(x', y') \in \mathbb{R}^2$ such that

$$2xx' + 2yy' = 0.$$

But this equation is exactly what makes (x', y') (viewed as a vector rooted at (x, y)) a tangent to the circle! So, the fibers of $X(A) \to X(A/I)$ are just tangent spaces of X.

5.5 Kahler differentials

For the more serious study of derivations, it's helpful to know how to compute them.

It turns out that the one kind of mathematics people are good at is linear algebra. Modules are pretty close to linear algebra; derivations are pretty far. But, by a miracle, we can turn problems about derivations into problems about modules, with the following trick.

Theorem 5.5.1 (Kahler differentials)**.** *Let R be a ring. Then, there exists an R-module Ω^1_R, called the* Kahler differentials *of R, with the following property: for every R-module M, there is a ("functorial in M," whatever that means...) bijection*

$$\mathrm{Der}(R, M) \simeq \mathrm{Hom}(\Omega^1_R, M).$$

More precisely, there exists a 'universal' derivation $d : R \to \Omega^1_R$, such that, for any R-module M, every M-valued derivation can be written uniquely as a composition

$$R \xrightarrow{d} \Omega^1_R \xrightarrow{\varphi} M,$$

for $\varphi : \Omega^1_R \to M$ some R-module homomorphism.

Example 5.5.2. Let $R = \mathbb{Z}[x]$, thinking of R as the affine line. Then

$$\Omega^1_R = \mathbb{Z}[x]dx,$$

by which I mean Ω^1_R is the free $\mathbb{Z}[x]$-module of rank 1 generated by a formal symbol dx.

The universal derivation

$$d : \mathbb{Z}[x] \to \mathbb{Z}[x]dx$$

sends $f(x)$ to $d(f(x)) := f'(x)dx$.

Example 5.5.3. Let $R = \mathbb{Z}[x, y]/(x^2 + y^2 = 1)$ be the unit circle. Then

$$\Omega^1_R = \langle dx, dy \mid 2xdx + 2ydy = 0\rangle .$$

Here, we mean that Ω^1_R is the R-module generated by two abstract symbols dx and dy, modulo the one relation that

$$2xdx + 2ydy = 0. \tag{5.5.4}$$

Intuitively, you can think of dx and dy as being infinitesimally small quantities, and the relation (5.5.4) (which we also encountered in our discussion of high-dimensional Hensel's lemma last time) is exactly the relation needed for $(x + dx, y + dy)$ to still lie on the unit circle, given that (x, y) does. Thus Ω^1_R captures exactly the sort of algebra we were doing in high dimensional Hensel's lemma.

Proof. We leave this as an exercise to the reader. $\square$

Remark 5.5.5. We have now seen that derivations control the non-empty fibers of $X(A) \to X(A/I)$. Perhaps more surprisingly, derivations *also* control the surjectivity of $X(A) \to X(A/I)$. This is because of a general fact called *cohomology*: in a mathematical situation where you have some object which is not unique, but is unique up to some kind of understandable transformation (in our case, unique up to a derivation), then whether or not exists is usually controlled by some kind of 'cohomology' group. But to do this, one would need to know about something called the 'cotangent complex,' a certain enhancement of Ω^1_R. It turns out that, for any $x \in X(A/I)$, one can construct a class $\theta_x \in \mathrm{Ext}^1_R(\mathbb{L}_R, I)$, which vanishes if and only if x lies in the image of $X(A) \to X(A/I)$. The construction of θ_x depends a lot on how one defines the cotangent complex; I'm not sure I can say more without needing to define $\mathbb{L}_R$ in some way.